

Smart Waste Classification System — UI Mockups

Visual implementation reference. Values shown are illustrative and must be replaced by actual model results.

Login

WASTE AI

Smart Waste Classification

Email

demo@example.com

Password

.....

Sign In

Deep Learning • CNN • MobileNetV2

WASTE AI

Deep Learning System

Prediction

Dashboard

Dataset

Model Performance

History

About

Training

Dashboard

Local AI Demo

Classes

6

Waste categories

Test Accuracy

94.2%

Example UI value

F1 Score

0.936

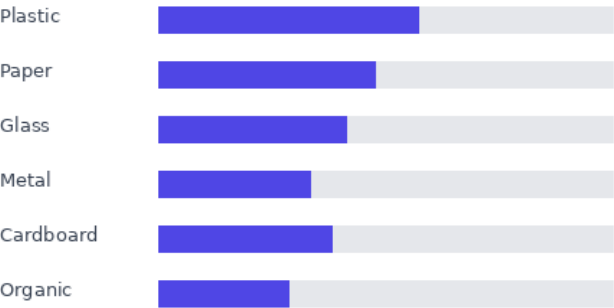
Macro F1

Model

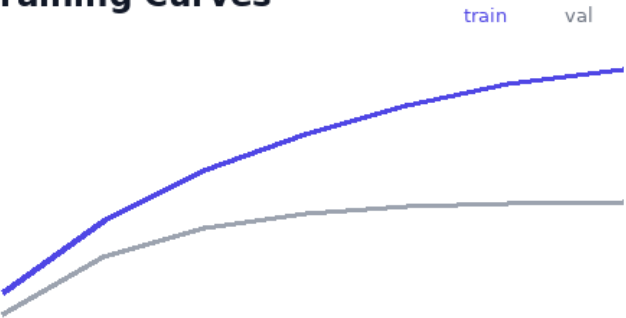
MobileNetV2

v1.0

Class Distribution



Training Curves



Recent Predictions

plastic_bottle.jpg	Plastic	96.4%
paper.jpg	Paper	91.2%
can.png	Metal	88.7%

Waste Prediction

Local AI Demo

Upload Waste Image

Drop image here

JPG / PNG / WEBP

Choose Image

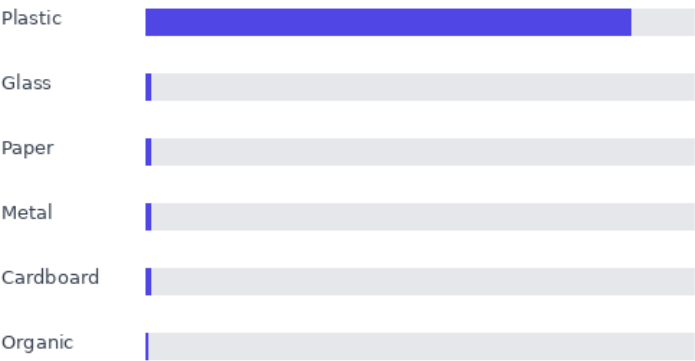
Predict Waste

Prediction Result

PLASTIC

Confidence: 96.4%

Class probabilities

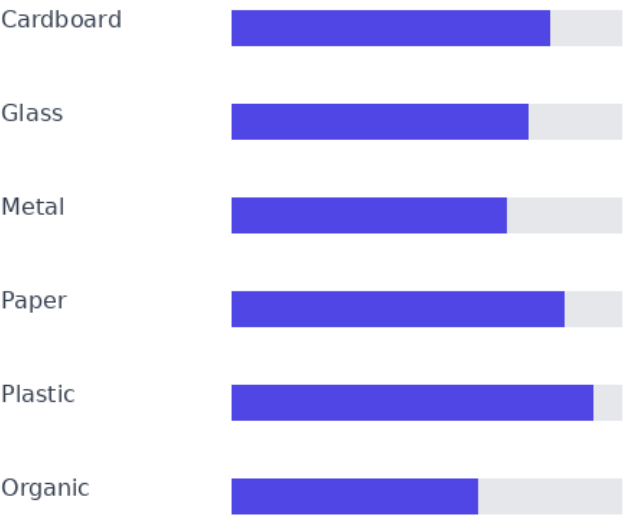


Model: MobileNetV2 v1.0

Waste Image Dataset

Classes: 6 • Train: 70% • Validation: 15% • Test: 15%

Class Distribution



Dataset Information

Source: documented public dataset

Total images: actual dataset count

Image formats: JPG / PNG

Duplicates: checked before split

Class mapping: class_names.json

Test set: never used for training

Model Performance

Local AI Demo

MobileNetV2 • v1.0

Input: 224×224 • Framework: TensorFlow/Keras

Accuracy

94.2%

Test set

Precision

93.8%

Test set

Recall

93.5%

Test set

F1

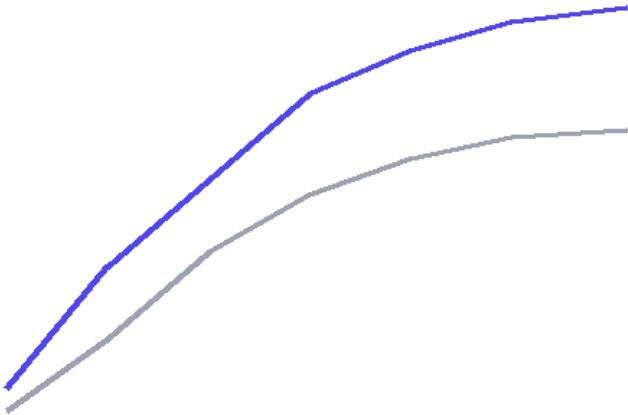
93.6%

Test set

Confusion Matrix

90	11	12	13	14	15
11	90	13	14	15	16
12	13	90	15	16	17
13	14	15	90	17	18
14	15	16	17	90	19
15	16	17	18	19	90

Training / Validation



Prediction History

Local AI Demo

Recent Predictions

Time	Filename	Prediction	Confidence	Model
21:32	plastic_bottle.jpg	Plastic	96.4%	v1.0
21:29	glass_jar.png	Glass	92.1%	v1.0
21:25	can.jpg	Metal	88.7%	v1.0
21:18	paper.png	Paper	91.2%	v1.0

WASTE AI

Deep Learning System

Prediction

Dashboard

Dataset

Model Performance

History

About

Training

About Project

Local AI Demo

Smart Waste Classification

Problem

Automatically classify waste images to support faster digital waste segregation.

Deep Learning

Custom CNN baseline + MobileNetV2 transfer learning.

Classes

Cardboard • Glass • Metal • Paper • Plastic • Organic

Limitations

Predictions depend on image quality and dataset coverage. The model may perform poorly on unseen waste types or unusual backgrounds.

Training Lab

Local AI Demo

Training Configuration

Architecture

MobileNetV2

Epochs

20

Batch Size

16

Learning Rate

0.001

Image Size

224 × 224

Start Training

Training Progress

Epoch 12 / 20



Train Accuracy: 93.1%

Validation Accuracy: 91.8%

Validation Loss: 0.31

Best checkpoint: epoch 11

Training runs should save history, metrics and the best model artifact.

System Settings

Active Model	MobileNetV2 v1.0
Input Size	224 × 224
Max Upload	10 MB
API URL	http://localhost:8000/api/v1
Prediction History	Enabled